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Shale-Drilling Fluid Interactions: An Experimental Study to Assess the Impact on Shale Inhibition

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Abstract

The objective of this paper is to discuss shale-fluid interactions while drilling clay-rich shale formations. Shale swelling, dispersion and sloughing are frequently encountered problems that may lead to excessive non-productive time. Comprehensive understanding of the behavior of the shale sample helps in mitigating shale drilling problems.

To understand the behavior of a shale sample when exposed to fluids, different types of characterization tests have been conducted starting with mineralogy identification using X-ray diffraction. Testing campaign also included capillary suction time test as well as utilizing nuclear magnetic resonance to study diffusion behavior in shale matrix with and without the presence of shale inhibitors.

After carrying out X-ray diffraction and thermogravimetric analyses on shale samples to determine clay content and organic carbon content, a series of shale inhibition and water imbibition tests were conducted using non-inhibitive and inhibitive fluids such as polyamine and potassium chloride brines.

This paper provides a review of various factors that influence shale-drilling fluid interactions. It presents a suggested scheme to assess the reactivity of shale in presence of different fluid types. Preliminary results are presented here while it is recommended to conduct further study to validate the use of nuclear magnetic resonance under different conditions to monitor the imbibition/diffusion of water molecules into shale matrix.

Introduction

To understand how shale rock interact with fluids, it's essential to conduct a comprehensive characterization starting with analyzing their pore structure, mineralogy, and elemental composition using techniques like X-ray diffraction (XRD), X-ray fluorescence (XRF), and Quantitative Evaluation of Minerals by Scanning Electron Microscopy (QEMSCAN) as well as Thermogravimetric Analysis (TGA) that is used to determine the total carbon content. That might also extend to imaging techniques like micro-computed tomography (micro-CT) that help identify micro-fractures and other features, while thin section analysis and Secondary Electron Microscopy (SEM) help confirm mineralogy, crystal structure, and visualize layering and clay sheet shape.

After shale rock characterization comes reactivity testing, such as cation exchange capacity (CEC) and capillary suction time (CST) to predict how clay may respond to different fluids. Adsorption isotherms are also used to predict water uptake, which affects the distance between clay sheets, i.e. shale swelling.

Each shale sample contains a percentage of one or more clay types, along with quartz and other minerals. There are five main clay groups, with kaolinite, smectite, and illite being the most relevant to drilling activities. Clay structure consists of sheets with planes of atoms, which can be tetrahedral or octahedral. The clay should be neutral to avoid interactions with water or water-based muds. If not, interlayer materials neutralize the clay. The mineralogy of the shale sample determines its response to water. Understanding adsorption of water molecules onto clay surfaces is crucial to explaining the clay hydration process. Adsorption is the adhesion of atoms, ions, and molecules to a surface, and one or more layers of hydrate can adhere to surfaces with bond deficiency.

Shale Inhibitors may be organic or inorganic. organic shale inhibitors such as amine-based additives are very effective on shale with high cationic exchange capacity. On the other hand, salts such as calcium chloride, sodium chloride and potassium chloride are only effective as long as the water-based drilling fluids containing these salts are in contact with the clay. Though, since the salts are exhausted or displaced by fresh water, the clay will again hydrate and swell, destabilizing the drilled formation. Salts delay the swelling of the clay through a variety of mechanisms. The cation exchange reaction decreases the amount of water that can be absorbed by hydratable cations on the surface of the swelling clay. Also, a large amount of salts have been used to increase the ionic concentration of the water phase in order to retard osmotic hydration.

Diffusion of aqueous fluids into shale pore system

Comprehensive understanding of the diffusion process into shale matrix is important in order to tackle shale-fluid interactions. It involves the mass transfer of fluid molecules into micro, meso as well as macro pores. Fick's first law defines the diffusion flux as (Karger and Freude, 2002):

$$j = -D_T \frac{\partial c}{\partial z}$$

Where j is the diffusion flux, D_T is the transport diffusivity, c is the concentration of the diffusing species and z is the space coordinate in the direction of the concentration gradient. The diffusion process is monitored using mean square displacement and pore sizes.

Nuclear magnetic resonance, NMR, is a useful tool that can be used to study the diffusivity of different types of fluids into porous media. Relaxation times (T_1 and T_2) and diffusivity measurements can be used to describe the diffusion process and evaluate the diffusivity of the imbibed fluid. The diffusivity depends on several factors such as fluid viscosity and pore size. After understanding the diffusion process, ways to reduce the diffusivity of water into shale matrix can be recommended and, as a result, negative impact of shale-fluid interactions can be minimized. In literature, there exist several studies to analyze shale porosity and the imbibition of different fluids into shale pore network. They are carried out using NMR to study the fluid flow through porous media and the effect of movable fines on oil production. (Appel, 2004; Washburn et. al., 2014; Xu et al., 2015)

T_2 relaxation time can be calculated using the following equation (Gomaa et. al., 2014):

$$\frac{1}{T_2} = \rho_2 \frac{S}{V} + \frac{1}{T_2^{bulk}} + \frac{1}{T_2^{diff}}$$

Where ρ_2 is the surface relaxivity, S is the surface area of the pores and V is the volume of the pores. The first term on the right-hand side is the major contributing term to T_2 values while the other two terms can be neglected for shale formations given that they have long bulk relaxation times and assuming the internal gradient is low.

Chitale et. al. (2003) investigated the clay typing in formation evaluation using NMR. They utilized the NMR T_2 distribution to study the clay-bound water in water saturated as well as vaporized montmorillonite clay. It was observed that T_2 distribution was uni-modal for hydrated clay samples at less than the critical saturation, 500 mg of water per gram of dry clay. However, the T_2 exhibited bi-modal distribution at higher saturation values.

Meng et. al. (2015) investigated the mechanism by which spontaneous imbibition takes place in shale reservoirs. Their work aim was to explain preferential imbibition of fluids into small and large pores for two scenarios: co-current and countercurrent. Unlike the case of sandstone and volcanic rock, they found preferential imbibition in the case of shale where micro pores have the ability to imbibe more than large pores.

Related to diffusion and fluid flow through shale pore network, water activity, or salinity, of the mud compared to that of the pore fluid is one of the factors to be considered in controlling shale reactivity. For a drilling fluid with a high-water activity, i.e. low salinity, water will be driven from the drilling fluid into the shale formation with the aid of osmosis phenomenon. As a result, pore pressure increases, confining pressure decreases, shale starts to swell and wellbore will be destabilized. Because water activity is inversely related to salinity, raising salinity of drilling mud can enhance wellbore stability (Chenevert, 1970). However, previous experimental work showed that increasing salinity above an optimum value can have adverse effects due to the possibility of fracturing the shale and thus can cause wellbore instability. This is because too high salinity will cause pore pressure to drop down significantly due to water movement from formation into the wellbore leading to the formation of cracks and fissures within the shale matrix.

Sometimes mud weight is increased for mechanical stabilization of the borehole wall. Because the effect of high mud weight is time-dependent, it is usually increased step-wise, e.g. mud weight is increased by 0.5 ppg for every step. The mud weight is increased until the weight that is capable of stabilizing the shale is reached. Breakout widths can be significantly reduced when the mud weight is increased for a constant mud activity (Zoback, 2007). Raising mud weights can stabilize wellbore by reducing the wellbore stresses below the shale strength. However, excessive increases in the mud weight can result in low rate of penetration and can create induced lost circulation problems. Therefore, mud weight should be optimized to the minimum that is just sufficient to achieve wellbore stability without inducing mud losses or causing other drilling problems (Steiger and Leung, 1992)

An equation was developed by Mody and Hale (1993) to determine the pore pressure increase as a function of water activity. Parameters considered in this equation include: membrane efficiency, temperature, molar volume of water and the activities of both the pore fluid and mud. However, time was not considered although shale instabilities are dependent on time in many cases. Time was considered as a factor in other models. These models allow the estimation of the time duration that a mud with a known activity can support the wellbore formation with a known pore fluid activity before failure. As a result, plot of failure vs. time can be generated and based on these plots, mud chemistry can be adjusted to enhance the wellbore stability.

Cohesive forces are the forces that keep the rock particles sticking to each other. When the filtrate, i.e. water, penetrates shale rocks, cohesive forces will be broken, reduced or damaged. Thus, there will be a decrease in the shear stress that is required for failure to occur. This results in movement along the existing fracture or along the weak planes leading to hole enlargement problem. Furthermore, when the filtrate invades the shale and the osmotic swelling starts, the rock will be softened and an increase in the rock volume is noticed because osmosis triggers swelling of particle surface as well as the shale layers (Hayatdavoudi and Apande). Another effect of osmosis is the pressure created in the rock due to the flow of fluid into the shale matrix. This will weaken the cohesive structure further.

Interactions between the shale rock and the mud have various effects on different properties related to rock strength. The friction between the rock particles increases the amount of shear required to cause rock

failure. However, due to rock-fluid interactions, this amount of shear is significantly reduced making the rock more prone to shear failure. Amanullah (1993) discussed one factor that controls the cohesion other than the water content. Upon entering the invaded shale rock, filtrate reduces the surface free energy. This reduction causes the cohesion to decrease as well. Due to the reduction in both the friction between the rock particles and the inter-particle cohesion, the resistance of the rock to shear is reduced.

Rock-fluid interactions cause tensile strength to decrease and consequently, may lead to fracturing of borehole walls. Similarly, these interactions reduce the compressive strength of the rock. This is due to the effect of swelling and associated high pore pressure.

Another concept to be discussed in this context is the mud pressure penetration. Due to capillary suction effect of shale matrix, there will be mud pressure penetration into the shale rock. A small amount of filtrate can penetrate due to the low permeability and high capillary suction of shales. This pressure penetration causes the pore pressure to increase and destabilizes the wellbore (Riley et. al. 2012). The mud pressure penetration mechanism was supported by an experiment carried out by exposing one shale sample to water. The shale sample contained 4% smectite and 25% illite with a low cation exchange capacity (CEC) value of 3 meq/100 grams. After several days, instead of swelling or dispersing in water, the shale sample developed fractures along the bedding planes with widths ranging from 5 to 45 μm . As fractures get wider with time, wellbore instability will be triggered in this type of shales.

NMR has been used extensively to study fluid imbibition into shale and other types of rocks for different applications. This paper uses this tool to investigate the imbibition of different shale inhibitive and non-inhibitive fluids into shale pore system from drilling point of view. Therefore, this piece of work tries to look at NMR T2 distribution as a tool to study the diffusion and propagation of water and shale inhibitors in an attempt to better explain clay-fluid interactions in the micro level. Two shale core plugs with two different fluids were tested. As it is relatively a new idea, further and repeated testing is necessary to prove the reproducibility of the results introduced in the current work.

Two shale core plugs were acquired to conduct NMR T2 experiment. For both samples, the initial T2 distribution was generated before starting the saturation process with two different types of fluids: de-ionized water and 5% KCl solution. The sample, then, were saturated in the corresponding fluids for 24 and 72 hours and, after that, they were tested again for T2 distribution. This was followed by extended core plug saturating at high pressure of 1500 psi for an additional one week. The initial T2 distribution was compared to the consequent T2 distributions taken at different times to reveal the diffusion of the corresponding fluids into shale pore network.



Figure 1—NMR Apparatus



Figure 2—Pressure cell to saturate the NMR samples at high pressure

Results and Discussion

NMR relaxation time (T_2) test

By applying magnetic field gradient, the positions in space for one nucleus over a short period of time can be estimated. Based on the molecular positions at two different points in time can detect the displacement and, therefore, diffusion. In this study T_2 curves are used to study the diffusion of water and KCl brine into one shale sample.

Results of NMR T_2 distribution curves, [Figure 3](#) and [Figure 4](#), were analyzed in terms of cumulative porosity and incremental porosity. The ultimate values of cumulative porosity were compared to determine how much water imbibed into the shale pore network at different time intervals, [Table 1](#), [Table 2](#) and [Figure 5](#). On the other hand, the incremental porosity values were compared to investigate the preferential imbibition; i.e. whether fluids imbibe first in micro or large pores.

Table 1—Results of NMR T_2 cumulative porosities of shale core plug saturated with deionized water for different saturation time periods

Saturation time (days)	Cumulative Porosity (p.u.)	Apparent water imbibed (%)
0	0.7	0
1	0.95	35.71
4	1.08	54.29
11	1.13	61.43

Table 2—Results of NMR T_2 cumulative porosities of shale core plug saturated with 5% KCl solution for different saturation time periods

Saturation time (days)	Cumulative Porosity (p.u.)	Apparent water imbibed (%)
0	1	0
1	1.28	28
4	1.42	42

Saturation time (days)	Cumulative Porosity (p.u.)	Apparent water imbibed (%)
11	1.5	50

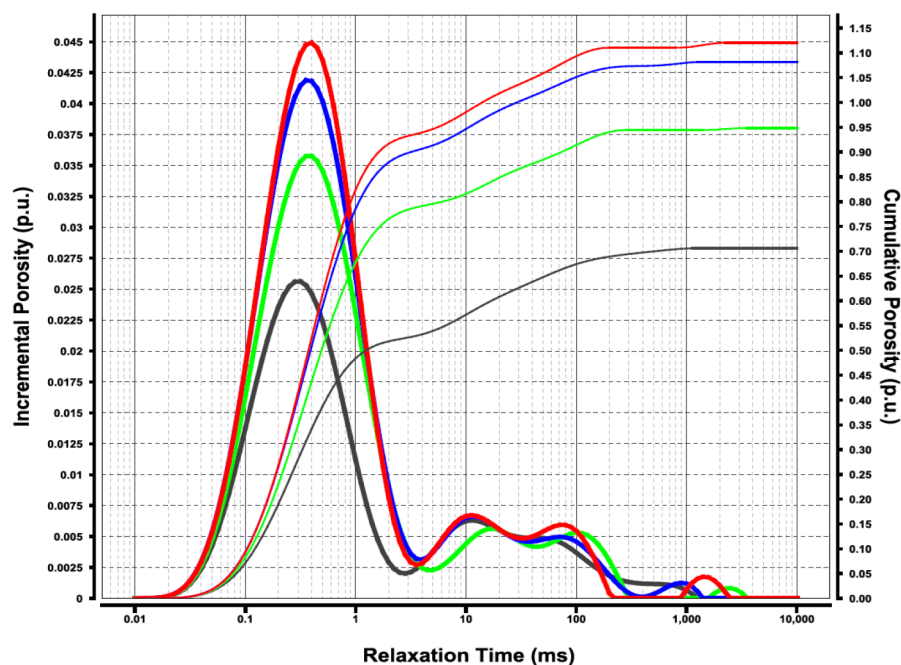


Figure 3—NMR T2 signal for shale saturated with water. Grey: dry sample. Yellow: after 24 hours of immersion. Blue: after additional 72 hours. Red: after additional 7 days at pressure of 1500 psi

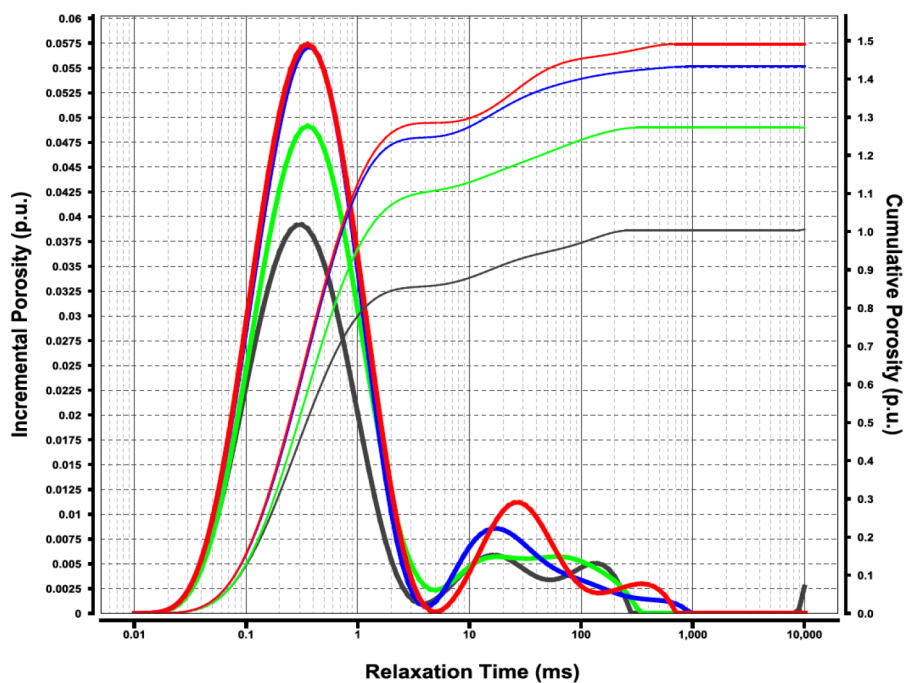


Figure 4—NMR T2 signal for shale saturated with KCl. Grey: dry sample. Yellow: after 24 hours of immersion. Blue: after additional 72 hours. Red: after additional 7 days at pressure of 1500 psi

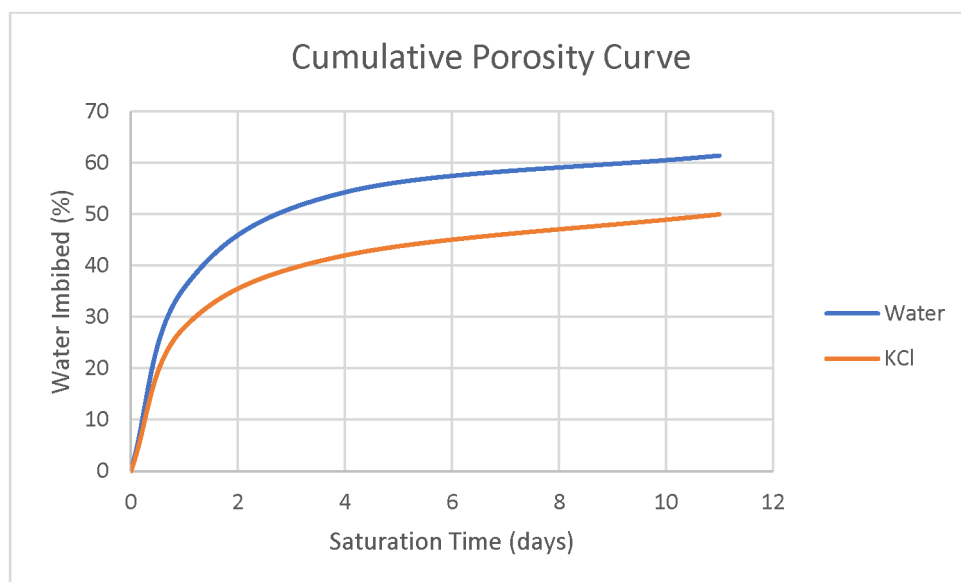


Figure 5—Comparison of percentage of water imbibed at the same conditions

The two shale core plug samples were acquired from the same well of sample C but at different depths of the same formation. The first plug sample was tested as a dry sample and the first T_2 distribution was plotted. The cumulative porosity curve showed a porosity value of 0.7 p.u. After saturating the sample in deionized water for 24 hours, there was a sharp increase in the cumulative porosity to 0.95 p.u. Upon further exposing to water for 72 hours and 1 week, the cumulative porosity continued to increase to 1.08 and 1.13 p.u. respectively. Assuming this increase is directly related to the amount of water imbibed into shale pore network, the amounts of water imbibed were determined for every step. The results showed that the water amount increased by 61.3% at the end of the 11th day. On the other hand, the second sample was tested as a dry sample showing cumulative porosity of 1 p.u. Although it had higher porosity initially, the amount of water imbibed after 24 hours was lower than that for the first plug sample. The same trend was observed after saturating the sample for extended periods of time: 72 hours and 1 week. This decrease in water imbibition might be attributed to the inhibition performance of KCl which results in smaller amount of water invading the shale pore network.

The other parameter analyzed is the incremental porosity. As it is clear from the T_2 distributions for both samples, the difference in the signal amplitude is much larger at smaller relaxation times where the micro-pores exist. This indicates the preferential imbibition to the micro-pores rather than macro-pores and micro-fractures. One explanation for this behavior is that the micro-pores are well inter-connected compared to the macro-pores. Therefore, the micro-pores are filled with water before the macro-pores.

This finding should be considered when developing new types of shale inhibitors. Shale inhibitors comprising nano-scale particles can plug those micro-pores and prevent further advancement of water. Also, amines and polyamines with their small particle sizes can result in high quality shale inhibition. It can also be observed that for the KCl brine case, higher amounts of water diffuse into the macro pores especially at high injection pressure. The peaks continue to rise as the saturation increases while it stopped increasing for micro pores. The observation suggests that adding another shale inhibitor that can plug the macro pores, such as silicates, can result in reduced amounts of water to invade the shale pore network.

A new generation of shale inhibitor may comprise both nano-particles to reduce the permeability by plugging micro-pores and, at the same time, functional groups such as polyamines and asphaltenes that showed good performance in terms of shale inhibition capability. The synergy of both mechanisms can provide shale inhibition with higher stability.

Conclusions

Understanding the diffusion process in shale matrix is crucial for managing shale-fluid interactions. This process involves the transfer of fluid molecules into micro, meso, and macro pores. Nuclear magnetic resonance (NMR) is a valuable tool for studying the diffusivity of fluids in porous media. By analyzing relaxation times (T1 and T2) and diffusivity measurements, researchers can gain insights into the diffusion process and evaluate the diffusivity of imbibed fluids.

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